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KNOWledge Nugget Of the Week

cognizant

Topic: Master Orchestrator — End-to-End Pipeline Controller

Scope:

The Master Orchestrator is the single entry point for the AI test automation platform. Instead of running each sub-framework individually (API / Swagger / Jira / Figma), the user drops input files into `inputs/<framework>/` and runs one command. The orchestrator auto-dispatches each subfolder to its framework via thin adapters, manages triage + retries on failure, applies LLM patches to broken test blocks where appropriate, narrowly re-runs only failed tests, aggregates self-healing telemetry, surgically pushes consolidated Pass/Fail to Zephyr, and prints one summary at the end. The subfolder name *is* the framework type — no classification, no LLM guessing. Execution is strictly serial in a fixed order (api → swagger → jira → figma) so upstream artefacts are available downstream.

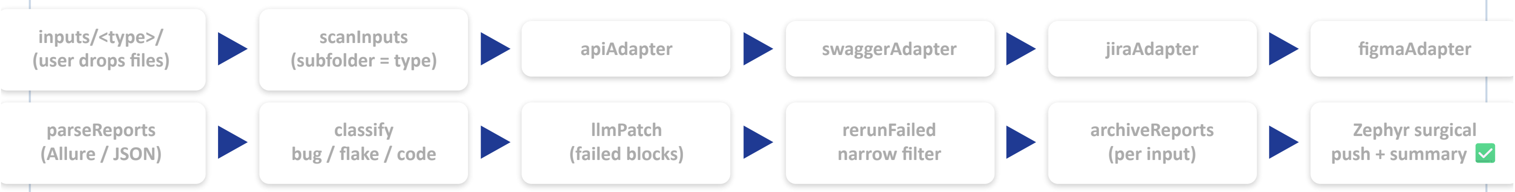
Key Features of the Master Orchestrator:

- **Folder-Driven Dispatch:** `inputs/api/` (*.json), `inputs/swagger/` (*.md + swagger.yaml), `inputs/jira/` (*.md), `inputs/figma/` (file JSON + frames PNGs) — subfolder = framework. Empty folders skipped silently.
- **Strict Serial Execution:** api → swagger → jira → figma (configurable via `config.FRAMEWORK_ORDER`); each framework's outputs feed the next when relevant.
- **Four Per-Framework Adapters:** `apiAdapter` spawns `run-tests.bat`; `swaggerAdapter` injects story into `swagger.yaml` first; `jiraAdapter` runs generation + automation back-to-back; `figmaAdapter` runs with `FIGMA_LOCAL_ONLY=1`.
- **Universal Triage:** `parseReports.js` normalises Allure (API + Swagger), Playwright JSON (Jira) and LLM JSON (Figma) into a unified `{passed[], failed[], errors[]}` shape; classifies each failure as real bug / test-code defect / transient flake.
- **LLM Surgical Patch:** `llmPatch.js` sends failing test block + error + page object to Claude (Sonnet 4.6); returns a replacement of that block only — never the whole file. Enabled per-framework via `LLM_PATCH_ENABLED` (Swagger + Jira on; API off — failures are data/auth, re-run suffices).
- **Narrow Re-Run:** `rerunFailed.js` builds per-framework filters — Maven `-Dtest=Class#TC_001*+TC_005*`, Playwright `--grep "TC_001|TC_005"`. Up to `MAX_RETRY_ATTEMPTS = 2`; surgical Excel update after each retry.
- **Healing Bridge:** `healingBridge.js` aggregates per-framework `healing-summary.json` into one consolidated view; rendered via `node index.js --healing-report` at any time.
- **Per-Input Archive:** `archiveReports.js` moves the framework's raw report into `<framework>/runs/<inputBaseName>/` — every input has a permanent browsable record.
- **Surgical Zephyr Push:** `surgicalUpdateZephyr.js` consolidates results across all frameworks into one Zephyr update — only the cases that ran this session are touched.
- **CLI Flags:** no args = run every populated folder; `--framework=<type>` runs one only; `--healing-report` renders telemetry; `--help` for usage.

Why Use the Master Orchestrator?

Running four frameworks individually means four entry points, four sets of env vars, four report formats and four manual Zephyr updates. The orchestrator collapses that to one command. It also closes the feedback loop: failures are triaged, code defects are LLM-patched, only failed tests re-run, healing is aggregated, Zephyr is updated — all without human intervention between steps. The result is a single CI-ready pipeline that turns Jira stories, Figma mockups, REST catalogs and Swagger specs into validated, reported, healed test artefacts in one session. The folder-driven contract means onboarding a new story takes one file copy.

Master Orchestrator Architecture



Integration with Tooling, Config & CI/CD

Built on **Node.js + child_process.spawn** for subprocess management, **Anthropic Claude (Sonnet 4.6)** for the LLM-patch step, **ExcelJS** for consolidated summaries, **dotenv** for shared env. Consumes native reports of each sub-framework: **Allure** (API + Swagger), **Playwright JSON** (Jira), **LLM JSON** (Figma). Drives narrow re-runs via **Maven Surefire** (`-Dtest=...`) and **Playwright** (`--grep ...`). Aggregates the **self-healing** module's per-run telemetry through `healingBridge.js`. Knobs in `config.js`: `MAX_RETRY_ATTEMPTS`, `FRAMEWORK_ORDER`, `FRAMEWORK_TIMEOUT_MS`, `LLM_PATCH_ENABLED`, `ZEPHYR_PUSH_ENABLED`. NPM shortcuts: `npm run start | api | swagger | jira | figma | healing-report`. CI-ready — integrates with Jenkins / GitHub Actions / GitLab CI / Azure DevOps.

References: nodejs.org · docs.anthropic.com · playwright.dev · maven.apache.org · docs.qameta.io/allure · smartbear.com/zephyr

Courtesy: K Jyothi Prakash (2445084)

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